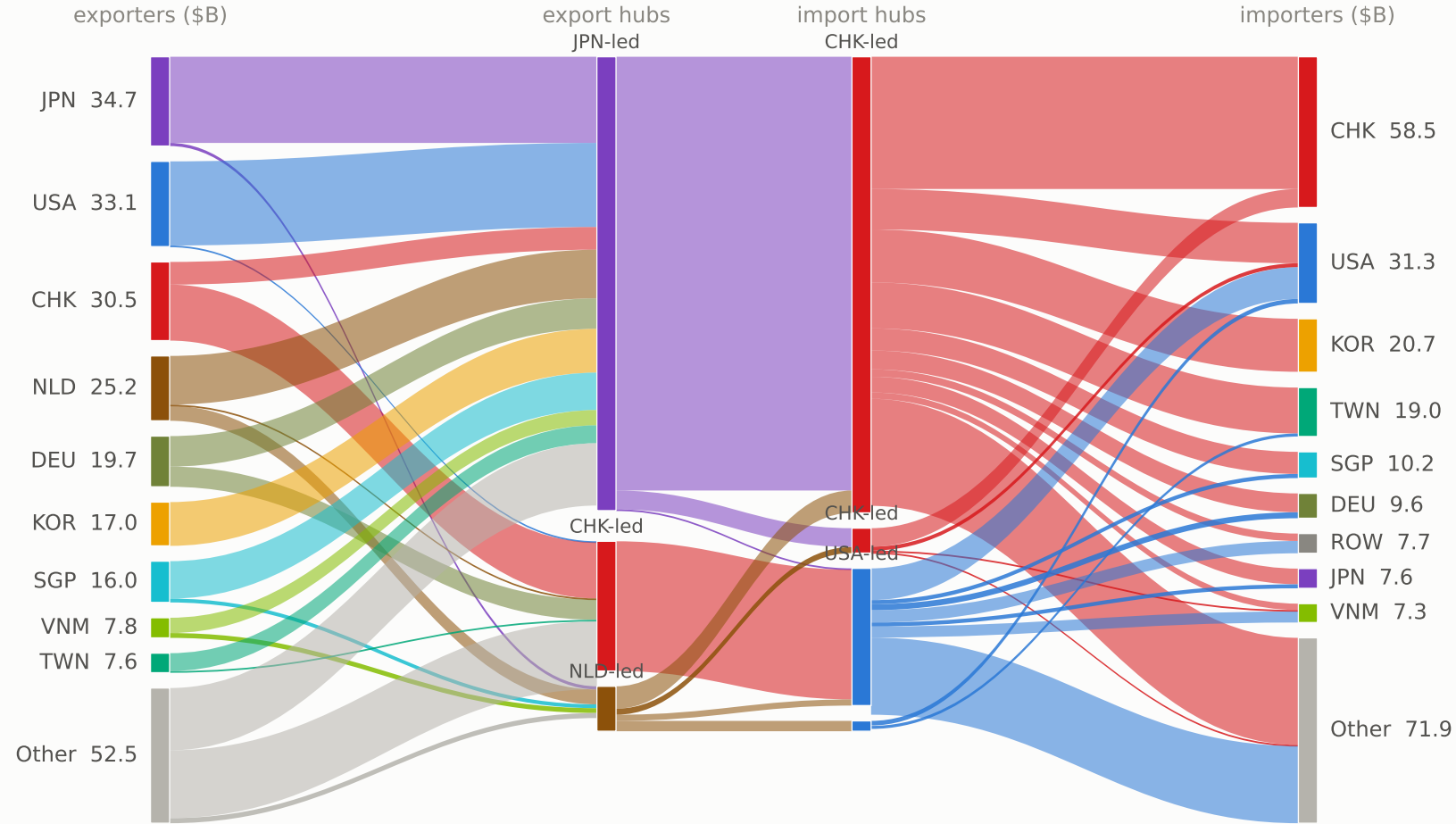


Stage 4 — Fab equipment [unrotated] (8486xx, metrology, fab plant) — 2024
 through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$244B



Hub decomposition from the factor model (UNROTATED spectral basis, hubs in descending eigenvalue order; constant-loading annual MFM 2020-24 (no monthly panel upstream)); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral.